

Name:-

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Assignment 1: Student Management Dashboard

```
English 2 > App.jsx > default
1  import StudentDashboard from "../components/StudentDashboard";
2
3  function App() {
4    return (
5      <div style={{ padding: "20px" }}>
6        <h1>Student Management Dashboard</h1>
7        <StudentDashboard />
8      </div>
9    );
10 }
11
12 export default App;
```

StudentDetails.jsx

English 2 > StudentDetails.jsx > default

```
1  import React from "react";
2
3  function StudentDetails({ students, deleteStudent }) {
4    return (
5      <div>
6        <h2>Student List</h2>
7
8        <ul>
9          {students.map((student, index) => (
10            <li key={index}>
11              {student}
12
13              <button
14                onClick={() => deleteStudent(index)}
15                style={{ marginLeft: "10px" }}
16              >
17                Delete
18              </button>
19            </li>
20          ))}
21        </ul>
22      </div>
23    );
24  }
25
26  export default React.memo(StudentDetails);
```

StudentDashboard.jsx

English 2 > StudentDashboard.jsx > default

```
1  import {
2    useState,
3    useEffect,
4    useRef,
5    useMemo,
6    useCallback,
7  } from "react";
8
9  import StudentDetails from "../StudentDetails";
10
11  function StudentDashboard() {
12
13    // useState
14    const [students, setStudents] = useState([
15      "Arjun",
16      "Rajesh",
17      "Harsha",
18    ]);
19
20    const [name, setName] = useState("");
21
22    // useRef
23    const inputRef = useRef();
24
25    // Auto focus
26    useEffect(() => {
27      inputRef.current.focus();
28    }, []);
29
30    // Add student
31    const addStudent = useCallback(() => {
32      if (name.trim() !== "") {
33        setStudents([...students, name]);
34        setName("");
35      }
36    }, [name, students]);
```

```

38 // Delete student
39 const deleteStudent = useCallback(
40   (index) => {
41     setStudents(
42       students.filter((_, i) => i !== index)
43     );
44   },
45   [students]
46 );
47
48 // Total students
49 const totalStudents = useMemo(() => {
50   return students.length;
51 }, [students]);
52
53 // Total characters
54 const totalCharacters = useMemo(() => {
55   return students.reduce(
56     (total, student) => total + student.length,
57     0
58   );
59 }, [students]);
60
61 return (
62   <div>
63     <h2>Student Dashboard</h2>
64
65     <input
66       ref={inputRef}
67       type="text"
68       placeholder="Enter Student Name"
69       value={name}
70       onChange={(e) => setName(e.target.value)}
71     />
72
73     <button onClick={addStudent}>
74       Add Student
75     </button>
76
77     <button
78       onClick={() => inputRef.current.focus()}
79     >
80       Focus Input
81     </button>
82
83     <p>Total Students: {totalStudents}</p>
84
85     <p>
86       Total Characters in Names: {totalCharacters}
87     </p>
88
89     <StudentDetails
90       students={students}
91       deleteStudent={deleteStudent}
92     />
93   </div>
94 );
95 }
96
97 export default StudentDashboard;

```

Output:-

Student Management Dashboard

Student Dashboard

Total Students: 3

Total Characters in Names: 17

Student List

Arjun	Delete
Rajesh	Delete
Harsha	Delete

Assignment 2: ProductList.jsx

```
ProductList.jsx > [🔍] default
1  import React from "react";
2
3  function ProductList({ products, removeProduct }) {
4    return (
5      <div>
6        <h2>Products</h2>
7
8        {products.map((product) => (
9          <div key={product.id}>
10            <p>
11              {product.name} - ₹{product.price}
12            </p>
13
14            <button
15              onClick={() => removeProduct(product.id)}
16            >
17              Remove
18            </button>
19          </div>
20        ))}
21      </div>
22    );
23  }
24
25  export default React.memo(ProductList);
```

App.jsx

App.jsx > default

```
1  import React, {
2    useState,
3    useMemo,
4    useRef,
5  } from "react";
6
7  import ProductList from "../ProductList";
8
9  function App() {
10
11    const [products, setProducts] = useState([
12      { id: 1, name: "Laptop", price: 60000 },
13      { id: 2, name: "Mobile", price: 25000 },
14      { id: 3, name: "Headphones", price: 4000 },
15      { id: 4, name: "Keyboard", price: 2000 },
16    ]);
17
18    const [search, setSearch] = useState("");
19
20    const searchInputRef = useRef();
21
22    const filteredProducts = useMemo(() => {
23      return products.filter((product) => {
24        product.name
25          .toLowerCase()
26          .includes(search.toLowerCase())
27      });
28    }, [products, search]);
29
30    const totalPrice = useMemo(() => {
31      return filteredProducts.reduce(
32        (total, product) =>
33          total + product.price,
34        0
35      );
36    }, [filteredProducts]);
```

```

37
38   const removeProduct = (id) => {
39     setProducts(
40       products.filter(
41         (product) => product.id !== id
42       )
43     );
44   };
45
46   return (
47     <div>
48       <h1>
49         Product Search and Price Calculator
50       </h1>
51
52       <input
53         ref={searchInputRef}
54         type="text"
55         placeholder="Search Product"
56         value={search}
57         onChange={(e) =>
58           setSearch(e.target.value)
59         }
60       />
61
62       <button
63         onClick={() =>
64           searchInputRef.current.focus()
65         }
66       >
67         Focus Search
68       </button>
69
70       <h3>Total Price: ₹{totalPrice}</h3>
71
72       <ProductList

```

```

70       <h3>Total Price: ₹{totalPrice}</h3>
71
72       <ProductList
73         products={filteredProducts}
74         removeProduct={removeProduct}
75       />
76     </div>
77   );
78 }
79
80 export default App;

```

Main.jsx


```
main.jsx
1  import React from "react";
2  import ReactDOM from "react-dom/client";
3  import App from "./App";
4
5  ReactDOM.createRoot(
6    document.getElementById("root")
7  ).render(
8    <React.StrictMode>
9      <App />
10    </React.StrictMode>
11  );
```

Output:-

Product Search and Price Calculator

Total Price: ₹91000

Products

Laptop - ₹60000

Mobile - ₹25000

Headphones - ₹4000

Keyboard - ₹2000